Assignment 4
Text and Sequences

Introduction

The purpose of this assignment was to apply neural networks to text data to examine how training sample size affects the choice between a learned embedding layer and a pretrained embedding. The dataset used in this assignment was the IMDB movie review dataset for sentiment classification.

There were two main approaches tested in this assignment. The first approach used an embedding layer that learned word representation directly from the training data. The second approach used pretrained word embeddings, which already contain relationships between words learned from a much larger dataset.

The main objective of this assignment was to determine how model performance changes as training sample size increases and to explain when one embedding method becomes more effective than the other.

Methodology

Learned Embedding Model

For the first set of tests, a neural network with a learned embedding layer was used. This model learns word representations directly from the training data. The model was made with an embedding layer followed by dense layers for classification. This approach depends entirely on the available training data to learn patterns.

Pretrained Embedding Model

For the second set of tests, a pretrained embedding approach was used. In this model word embeddings were initialized using pretrained weights rather than being learned from scratch. The pretrained embeddings give prior knowledge about word relationships. This helps the model perform better when training data is limited. The rest of the model structure stayed the same to make comparisons between the models.

Results

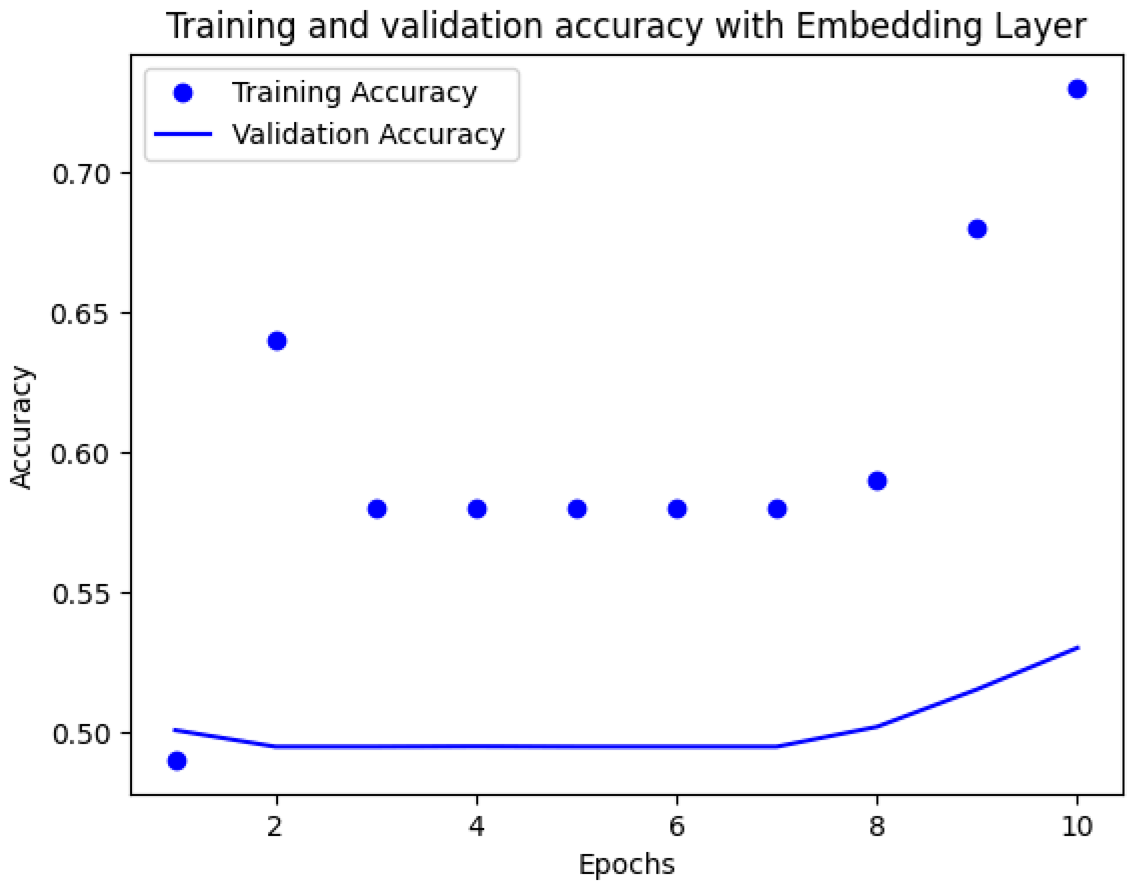
When the model used a learned embedding layer with only 100 training samples the accuracy was relatively low. This is because the model had very limited data to learn word relationships from scratch.

As the training size increased, the model showed steady improvement. With more data the model was able to learn patterns in the text and improve classification accuracy. At larger training samples the learned embedding model performed much better and became more competitive overall.

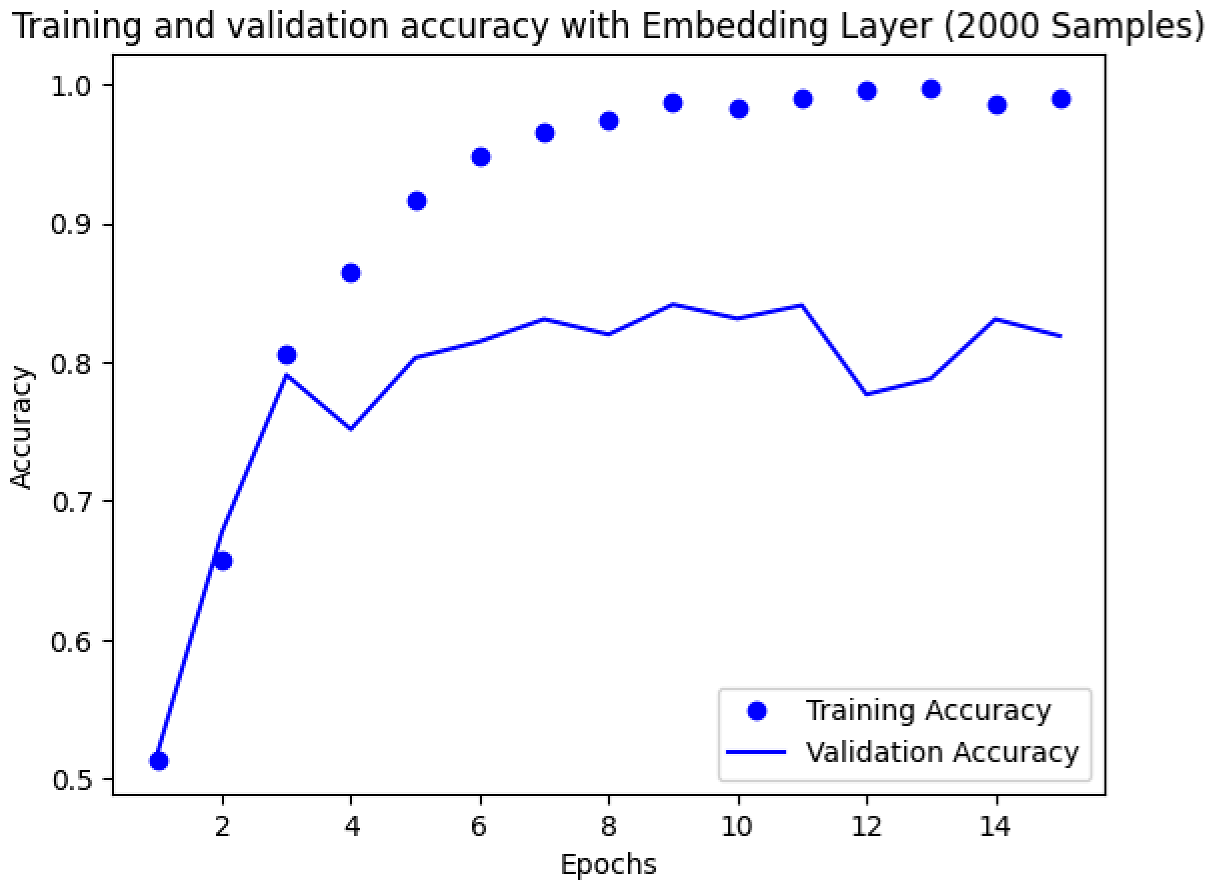
The pretrained embedding model performed similarly to the learned embedding model on only 100 training samples. This is because with only 100 samples it is very hard for a model to perform with very little to work with. As the training size increased the pretrained model also continued to do well. Since it already starts with useful embeddings, additional data does not impact performance as significantly.

Test	Model Type	Training Size	Validation Accuracy	Test Accuracy
1	Learned Embedding	100	0.51	0.522
2	Learned Embedding	500	0.731	0.73
3	Learned Embedding	2000	0.845	0.81
4	Pretrained Embedding	100	0.505	0.515
5	Pretrained Embedding	500	0.625	0.544
6	Pretrained Embedding	2000	0.625	0.61

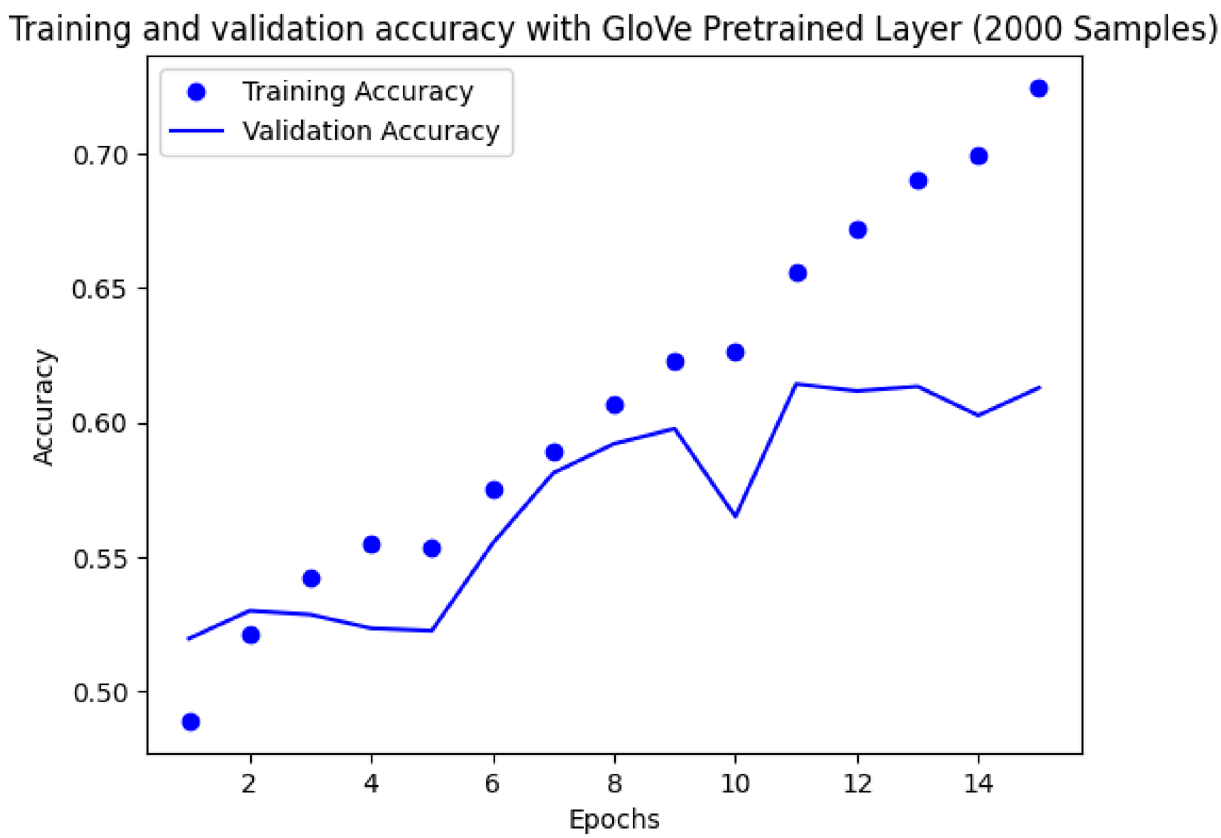
The table shows that the learned embedding model improves significantly as training samples increase. While the pretrained model shows more gradual improvement and lower overall performance at higher training samples.



In this figure, you can see that with only 100 training samples the learned embedding model only had about 55% validation accuracy.



In this figure, you can see the learned embedded model with 2000 training samples showing it improves significantly when having more training data reaching around 80%.



In this figure, You can see that with giving more samples to a pretrained GloVe model it only had around 60% validation accuracy.

Discussion

The results show a clear relationship between training sample size and embedding choice. When training data is small, a pretrained model may have the edge on a learned model because it does not rely heavily on the dataset to learn word meanings.

As training size increases the learned embedding model improves significantly starting at about 51% accuracy with 100 training samples up to 80% after getting 2000 training samples. This shows that with more data the model is able to learn more patterns and improve performance.

An important observation is that the pretrained model does not improve as much with more data. This is because it already starts with fixed word representations and cannot adapt as much to the specific dataset.

Another important observation is that the learned embedding model benefits more from increased training data, while the pretrained model remains more consistent but less adaptable.

Conclusion

In this assignment, neural networks were applied to text classification using both learned and pretrained embeddings. The results show training sample size has a significant impact on model performance.

The learned embedding model improved as more data was added, showing that it benefits from larger datasets. Both models performed similarly when the training data was limited, but the learned embedding model improved as more training data was available.

Overall, pretrained embeddings are a better choice for smaller datasets, while learned embeddings become more effective as the amount of training data increases.